Design and Integration of a Head Tracking System for Remote Camera/Gimbal Control on a UAV Drone

EENG 4990.001 Senior Design II

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KEYWORDS

- camera integration
- embedded systems
- gimbal control
- head tracking
- PCB design
- remote sensing
- wearable electronics

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Abstract

The project focused on developing a dedicated hardware platform capable of receiving head-tracking input, processing motion data, and transmitting control signals to the camera gimbal for pan and tilt movement. Electrical design considerations included signal routing, power distribution, filtering components, and connector integration, while mechanical design addressed mounting, alignment, and overall system fit. Prototype development and ground-based testing were completed to evaluate communication performance and system functionality. Results showed that the system successfully demonstrated head-tracking control and reliable wireless operation at distances of up to approximately 1 km under ground test conditions. Although full flight testing was not completed, the final prototype established a strong foundation for future integration into UAV-based remote sensing platforms. This work demonstrates the feasibility of a custom head-tracking control board and supports future improvements in stabilization, durability, and full aerial deployment.

Introduction

Head tracking systems are widely used in robotics, drones, teleoperation, virtual reality, and assistive systems because they provide a more natural and immersive method of control. Instead of relying entirely on joysticks or manual directional input, these systems convert the user's head movement into corresponding control signals. This allows the operator to direct a camera or gimbal system more intuitively and efficiently.

The goal of this project was to design and integrate a head tracking system capable of supporting a remote camera or gimbal platform. The system was intended to provide real-time directional control through head movement while returning a live video feed to the user through a headset. Achieving this required reliable electronics design, stable power regulation, practical PCB layout, and careful integration with both the camera and headset hardware.

Because the system is intended for a compact and mobile environment, the design had to emphasize low power consumption, hardware reliability, compact form factor, and ease of integration. This paper discusses the design process, implementation, testing, results, and future improvements of the head tracking system.

Problem Statement

Traditional remote camera control methods can be difficult to use when the operator must divide attention between navigation, observation, and directional adjustment. This is especially true in drone or remote vision applications where fast response and natural user interaction are important. A head tracking system improves usability by allowing the camera orientation to follow the operator's head motion.

However, implementing such a system introduces several engineering challenges. The system must provide stable power delivery, reliable communication, appropriate signal conditioning, and compact physical integration. It must also account for real-world issues such as control inversion, mounting constraints, and hardware debugging. The challenge of this project was therefore to develop a PCB-based hardware platform that could support these functions while remaining compact, reliable, and suitable for integration into a headset-camera system.

Objectives

The objectives of this project were to:

1. design a PCB for the head tracking system using Altium;
2. select and source components required for sensing, regulation, and communication;
3. develop a complete bill of materials for fabrication and assembly;
4. assemble and solder the PCB for functional verification;
5. validate power and signal behavior using laboratory test equipment;
6. integrate the board with the camera and headset system;
7. support real-time transmission and directional control capability;
8. correct control issues related to axis inversion; and
9. prepare the design for final system-level implementation.

Solution

Our solution is to use a custom PCB-based head-tracking gimbal system to control an FPV camera in real time. The system allows the camera to follow the natural head movements of the user, creating a more responsive and immersive remote viewing experience. The PCB serves as the main control platform and connects the microcontroller, wireless communication modules, power components, filtering capacitors, connectors, and servo motor outputs needed for operation. The system is powered by a lithium battery pack, allowing portable use and easier integration into a mobile platform. Head-tracking input is processed and transmitted wirelessly to the receiving side, where the servo motors drive the camera in pan and tilt directions. The system also includes custom mechanical components to support the camera and gimbal assembly while keeping the design compact and adaptable. Overall, the prototype functions as a head-tracking control system for remote sensing applications and demonstrated successful operation during ground testing, with wireless performance reaching distances of up to about 1 km. Although full flight testing was not completed, the design provides a strong foundation for future improvements in stabilization, integration, and expanded field use.

System Overview

The head tracking system was designed as an embedded control platform that converts user head motion into directional commands for a remote camera gimbal assembly. The system is composed of a sensing stage, a processing stage, a wireless communication stage, regulated power distribution, and a remote actuation stage. These subsystems operate together to provide real-time camera positioning and visual feedback for remote sensing applications.

The main controller used in the system is the Pro Mini ATmega328P 5V/16 MHz microcontroller, which serves as the central processing unit on the custom PCB. This microcontroller receives motion data from the inertial sensor, processes the orientation-related input, and generates the control data required for transmission to the remote platform. Its small size, low power requirements, and compatibility with the rest of the embedded design made it suitable for integration into the head-mounted control hardware.

Head motion is measured using the MPU6050 gyroscope and accelerometer sensor, which provides motion and orientation data to the microcontroller. The MPU6050 combines a 3-axis gyroscope and a 3-axis accelerometer in a single device, allowing the system to detect changes in head position and angular movement. This sensor data is used to determine the operator's motion and convert it into corresponding control signals for the remote camera system. The sensor communicates with the microcontroller through a digital interface and forms the primary input stage of the head tracking system.

Wireless transmission between the head-mounted unit and the remote camera platform is supported through an nRF24L01 transceiver module. This communication link allows processed motion data from the headset side to be sent to the receiving side of the system, where the camera or gimbal can respond accordingly. The wireless link is intended to support low-latency transfer of directional commands, which is important for natural and responsive camera movement.

The PCB also includes the required power regulation and filtering circuitry to support stable operation of the embedded components. Decoupling and bulk capacitors were included to reduce noise and maintain voltage stability for the microcontroller, sensor, and communication module. Proper power distribution was necessary to reduce electrical instability that could affect sensing accuracy, wireless transmission, or general system reliability.

On the receiver side, control signals are used to drive the gimbal mechanism so that its motion follows the user's head movement. This creates a closed operator feedback loop in which head orientation is measured by the MPU6050, processed by the ATmega328P, transmitted wirelessly through the nRF24L01, and then used to reposition the remote camera. A live digital camera feed is returned to the headset, allowing the operator to immediately observe the resulting view direction.

Overall, the system integrates embedded sensing, onboard signal processing, wireless communication, and remote actuation into a compact hardware platform. This architecture supports intuitive camera control for drone-based remote sensing and first responder applications.

Design

Design Overview

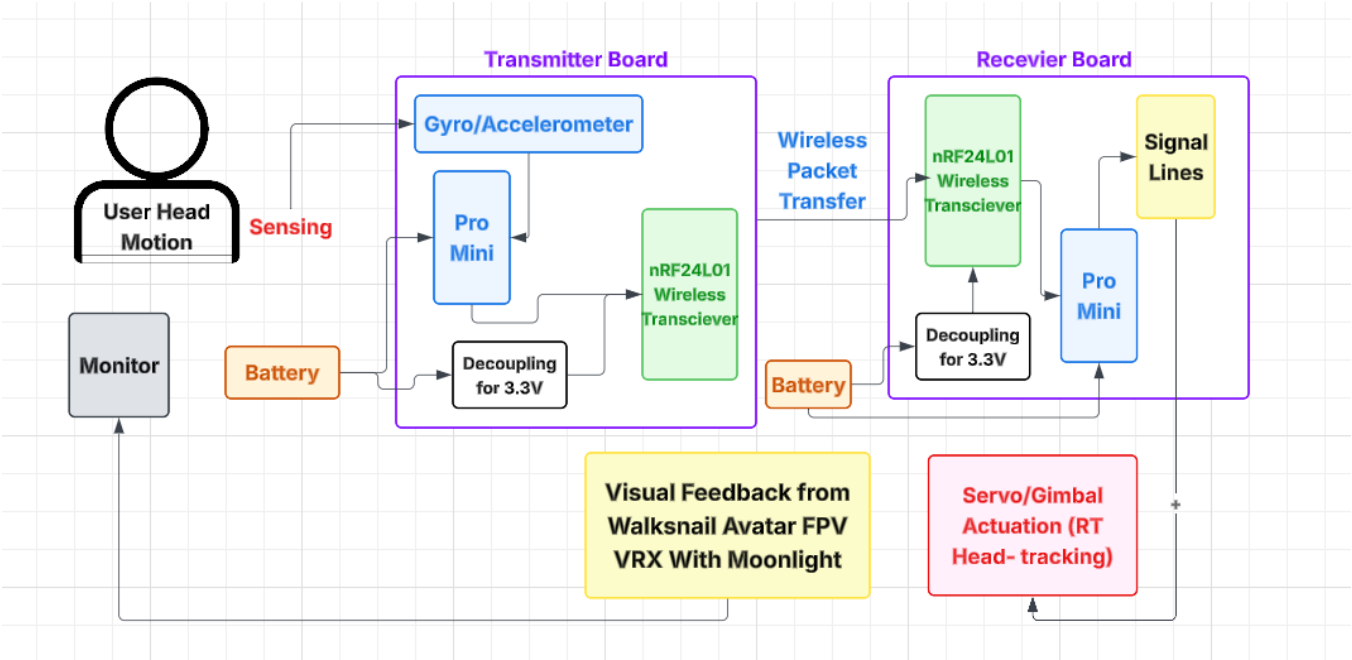


Figure 1: PCB Carrier Board Overview Block Diagram

System-level block diagram of the head tracking platform showing the user input, transmitter board, wireless packet-based communication link, receiver board, and visual feedback path. On the transmitter side, head motion is measured by the gyro/accelerometer and processed by the Pro Mini before being sent through the nRF24L01 wireless transceiver. The receiver board accepts the transmitted packets and outputs the control signals used for servo or gimbal actuation. The diagram also shows the battery-powered supply and decoupling required for stable 3.3 V operation, as well as the returned visual feedback from the FPV system to the user monitor.

Electrical Design

The electrical design focused on creating a reliable PCB capable of distributing power and supporting required interfaces for the head tracking system. Schematic capture was completed in Altium with attention given to connector placement, routing simplicity, electrical stability, and compact form factor. Proper filtering and decoupling were included to minimize noise and maintain stable supply rails.

Power integrity was an important consideration because instability in the supply could reduce sensing accuracy, affect communication performance, or create unreliable system behavior. Capacitor placement near active components and appropriate routing practices were therefore incorporated into the design.

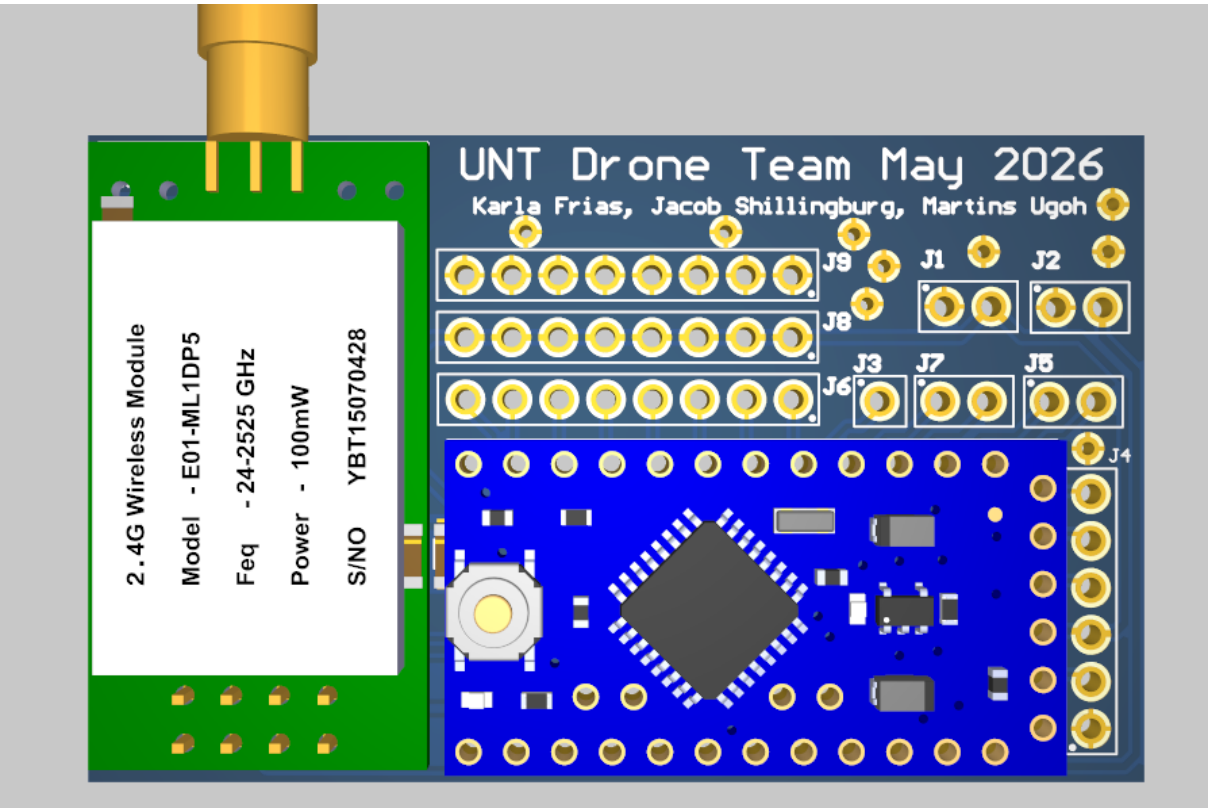


Figure 2: 3D PCB Carrier Board (Front)

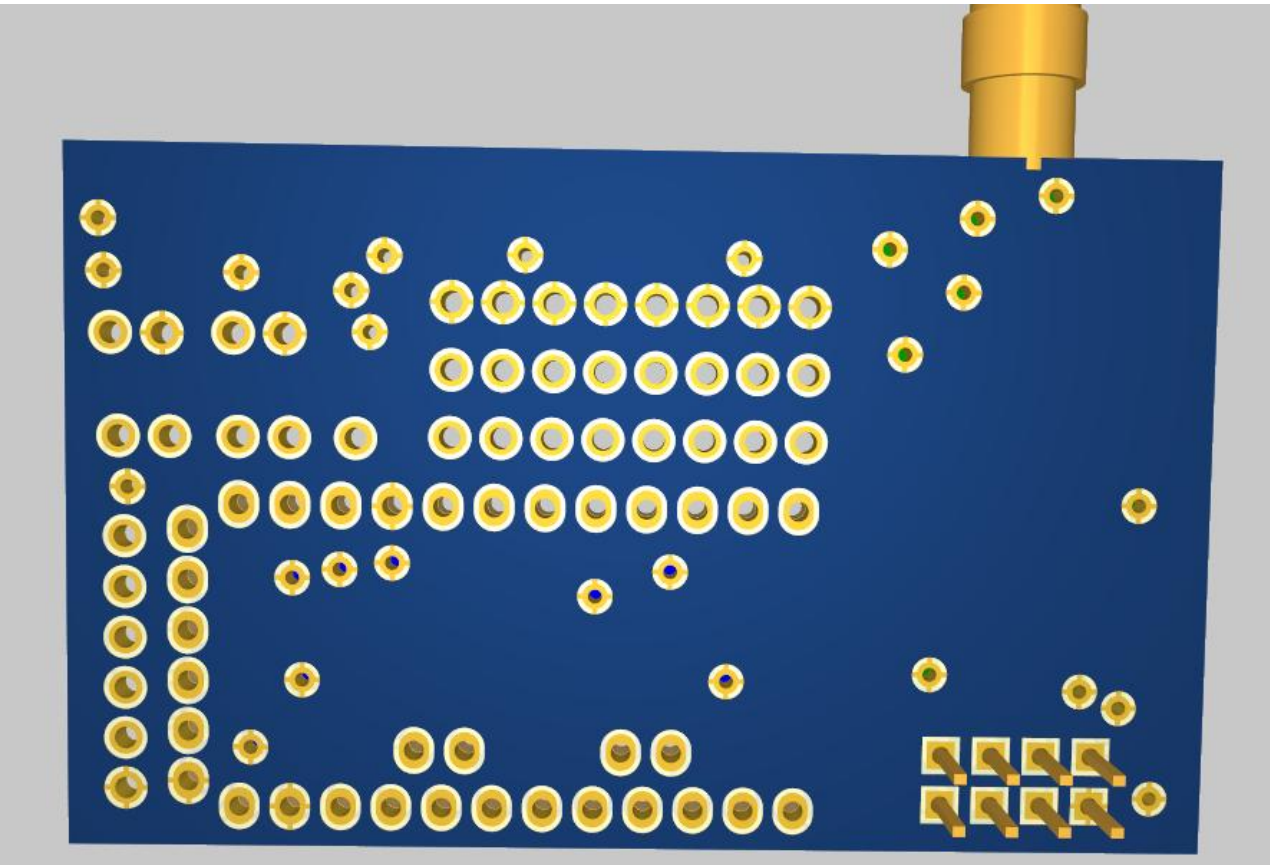


Figure 3: 3D PCB Carrier Board (Back)

The head tracking system was implemented on a two-layer printed circuit board designed in Altium Designer. After the schematic was completed, the components were assigned footprints and transferred into the PCB layout environment. The board was arranged to support the Pro Mini ATmega328P 5V/16 MHz, MPU6050 gyro/accelerometer, nRF24L01, and the required power regulation circuit in a compact carrier-board suitable for integration with the system.

The power section was designed around an AMS1117-3.3 voltage regulator to provide a stable 3.3 V supply. To support regulator stability and reduce electrical noise, the design used a combination of bulk and ceramic capacitors. C1 was 22 uF at the regulator input, and C5 was 10 uF at the regulator output for bulk filtering. The remaining capacitors, C2, C3, C4, C6, C7, and C8, were 0.1 uF, and were used for high-frequency decoupling and local noise suppression. This filtering approach helped improve voltage stability and support reliable operation of the embedded components.

Once the schematic and layout were finalized, routing was completed. Special attention was given to the placement of the voltage regulator, decoupling capacitors, and communication-related components so that power and signal paths could be kept as short and practical as possible. Design rule checks were then performed in Altium to verify spacing, connectivity, and board manufacturability before generating the final fabrication files. After fabrication, the board was soldered as part of the hardware implementation process. This included mounting the required passive and active components and preparing the board for electrical validation and integration

with the system. Following assembly, the board could then be used for validation of power delivery, signal behavior, and system integration with the camera and headset setup.

Component Selection

Component selection was based on electrical compatibility, compact integration, availability, and suitability for prototype testing. A major goal of the design was to use easily procurable parts, including components that could be sourced quickly from vendors on Amazon, so that assembly and testing could proceed without long procurement delays. This made the design more practical for rapid prototyping and reduced the risk of project delays caused by limited part availability.

The primary controller selected for the design was the Pro Mini ATmega328P 5V/16 MHz. This microcontroller was chosen because of its compact size, straightforward integration, and ability to interface with both the sensing and wireless communication hardware. It served as the main processing element of the transmitter board and handled the motion data used for head tracking control. The MPU6050 gyroscope and accelerometer was selected as the motion-sensing device. In this design, the MPU6050 was mounted directly on top of the Pro Mini, which meant it did not require additional PCB area on the carrier board. This arrangement helped reduce the effective space required by the sensing hardware and supported a more compact overall design. The sensor provided the motion information needed to determine the user's head orientation. Wireless communication was implemented using an nRF24L01 transceiver module. It was selected because it supports packet-based communication and is widely used in embedded control systems. The processed motion information was divided into two control components, pitch and yaw, before transmission. On the receiving side, these two data components were read and used by the servo system to drive gimbal pan and tilt motion. This allowed the head tracking system to convert user motion into separate directional control signals for remote actuation. Power regulation was supported using the AMS1117-3.3 voltage regulator, which provided the required 3.3 V supply for devices operating below the main input voltage. Capacitors were selected to support both regulator stability and noise filtering. Bulk capacitors were used to help smooth supply variations, while smaller ceramic capacitors were used for high-frequency decoupling. This combination provided a practical and effective filtering method for stable board operation.

In addition to the control electronics, extra design consideration was given to the return video system. More money was intentionally spent to obtain a digital video feed for the return path so the system could be tested more effectively at longer distances. This decision improved the ability to evaluate the platform under more realistic remote-operation conditions and supported clearer visual feedback during testing.

Overall, the selected components provided a practical balance of performance, compactness, cost, and availability. The use of readily available parts, compact sensor mounting, packet-based pitch and yaw transmission, and an upgraded digital return video system supported the development of a workable prototype for head tracking and remote camera control for first responders.

Mechanical Integration

Mechanical integration involved combining the electronics with the camera and headset structure. The design included the use of screws, M3 bolts, threads, and supporting mechanical features to secure the electronics and associated camera components in the final assembly. Mechanical constraints influenced the PCB dimensions, connector orientation, and overall system packaging.

This stage emphasized that successful embedded design requires coordination between electrical and mechanical considerations rather than treating them as separate tasks.

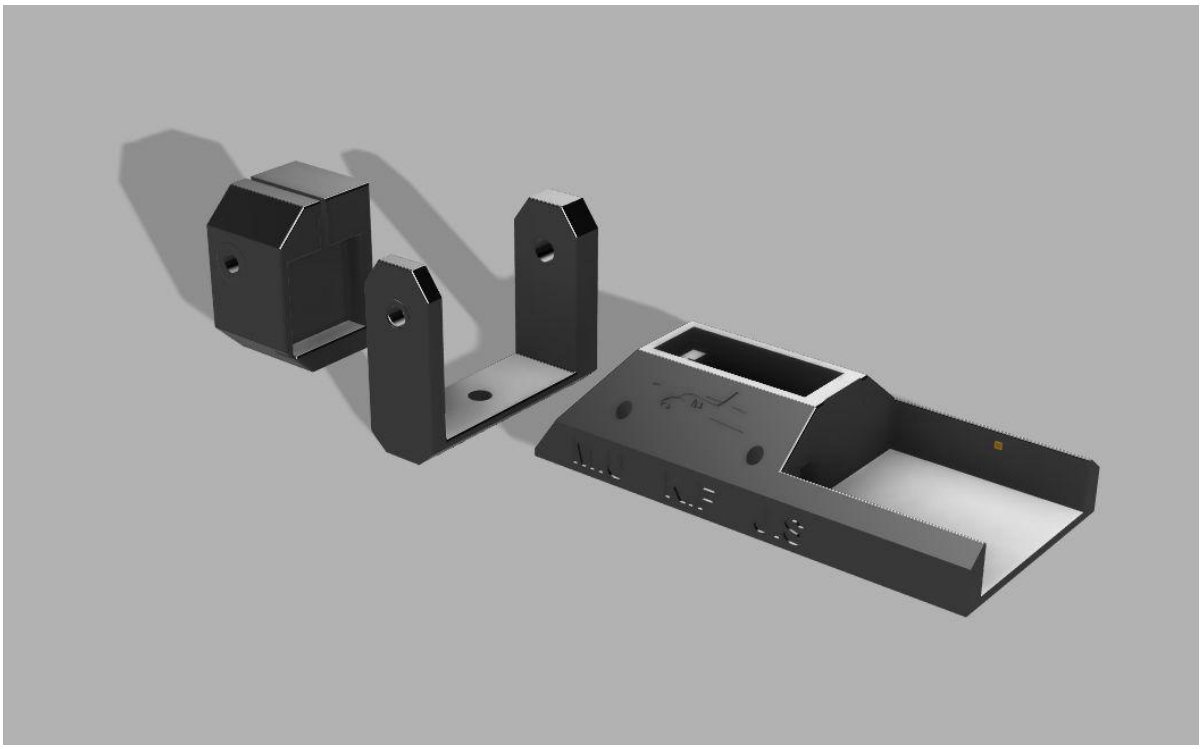


Figure 4: 3D Gimbal System Design

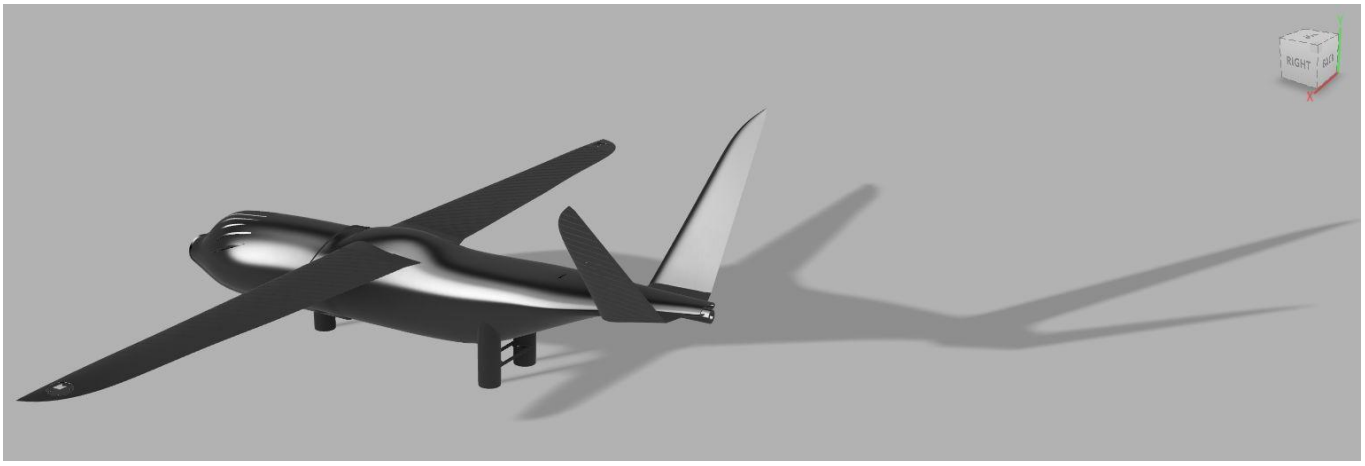
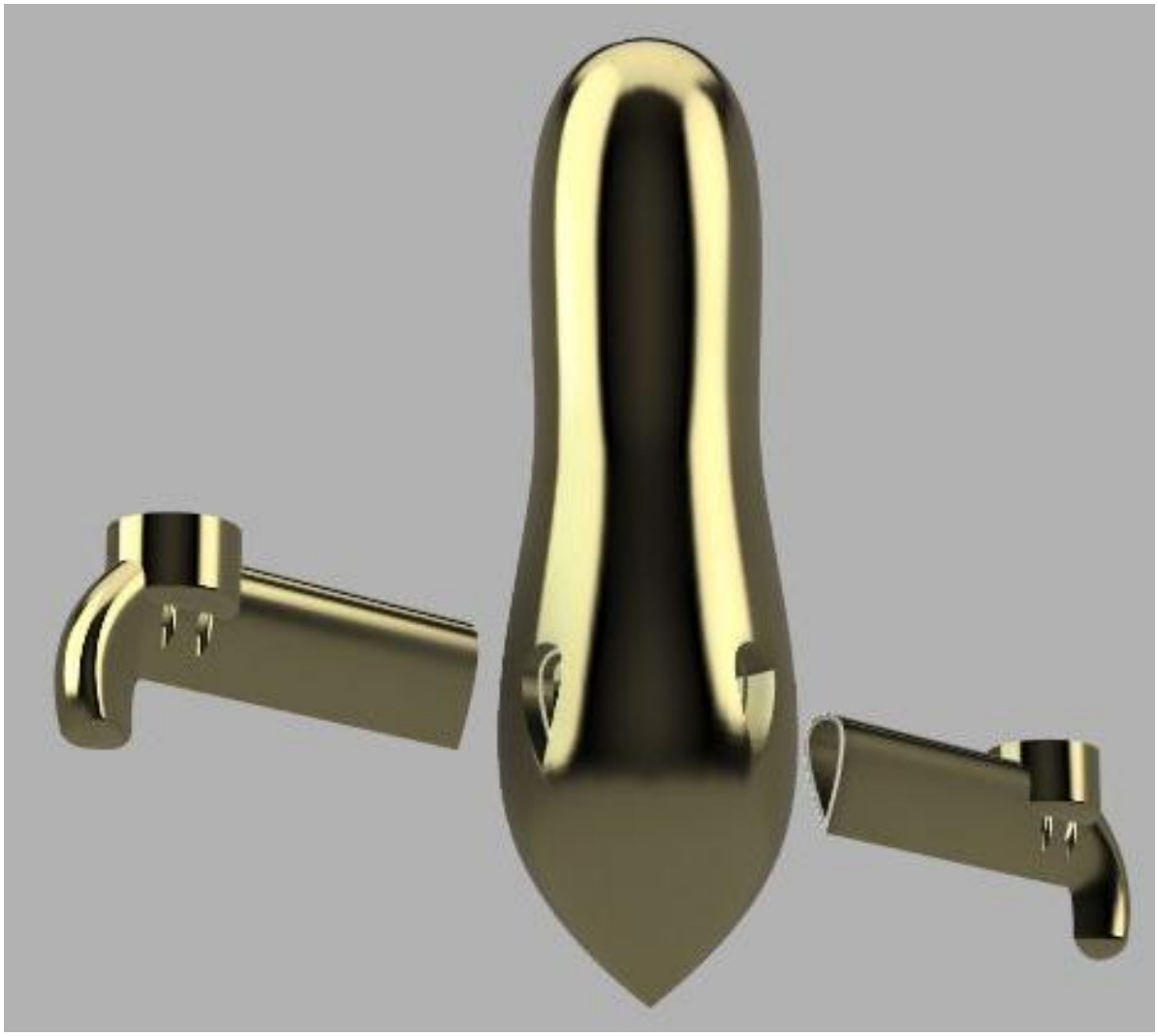


Figure 5: 3D V1 Drone Design



The mechanical design process focused on creating a practical structure for the carrier board, camera hardware, and related mounting features while maintaining a compact form factor suitable

for the intended drone platform. In addition to the head tracking hardware assembly, two drone configurations were also designed for test flights to evaluate system integration. Using Fusion 360, the mechanical components were modeled to support placement of the PCB, camera, and structural hardware within the available space.

The design also considered fastening and assembly requirements, including the use of screws, mounting holes, and support features needed to secure the board and camera components during testing and operation. Proper mechanical support was especially important because the system is intended for a moving platform, where vibration and repeated handling can affect reliability. By developing the structure in CAD before fabrication, the design process improved the likelihood of successful assembly and helped reduce integration issues during testing.

Implementation

Schematic and PCB Development

The implementation began with schematic development and PCB design in Altium. After the circuit was completed and checked, a bill of materials was prepared to support part procurement. Once components were obtained, the board was soldered and assembled.

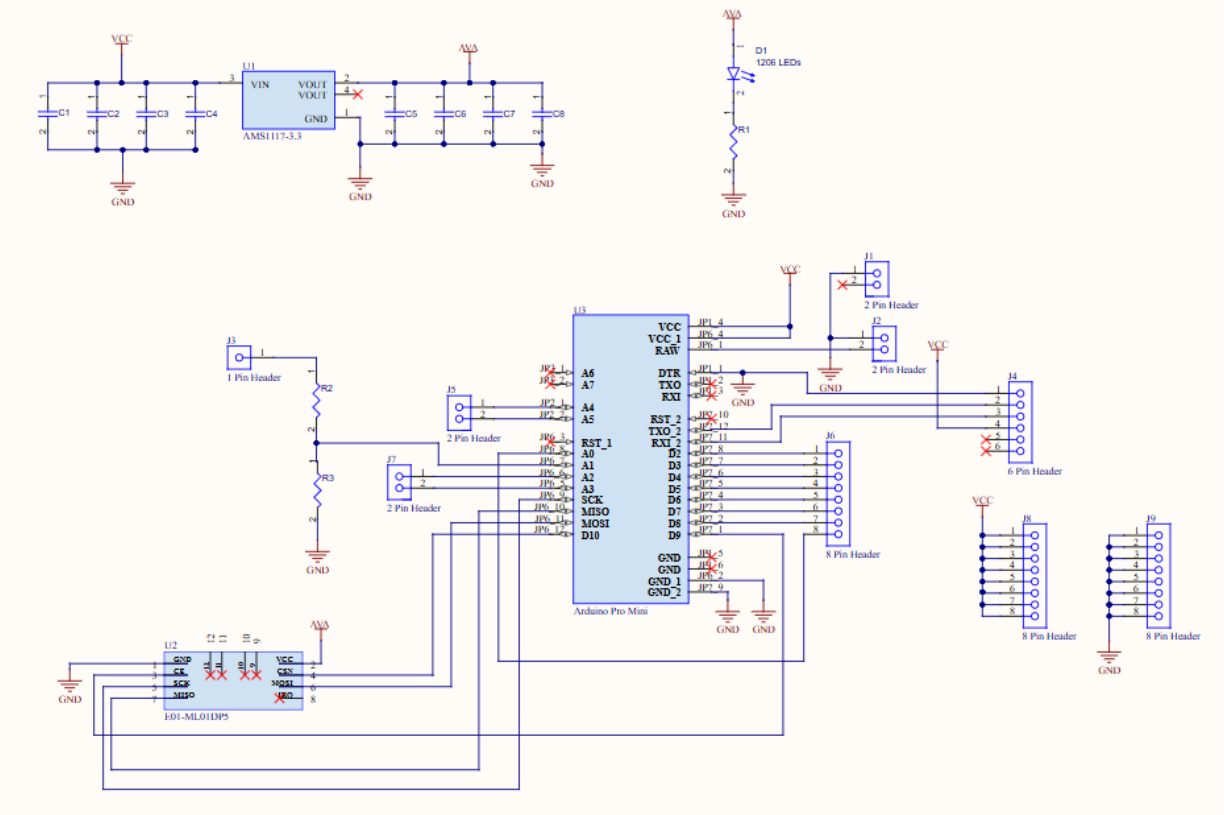


Figure 6: Altium Schematic for PCB Carrier Board

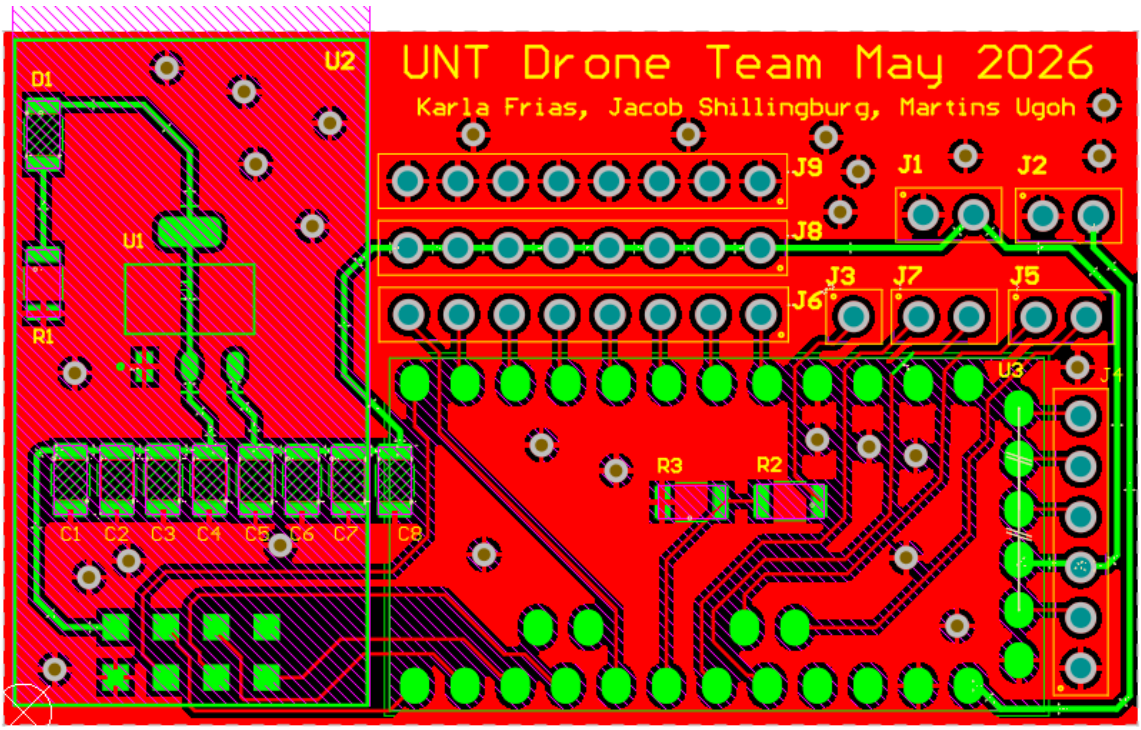


Figure 7: Altium Top Layer (GND and Signals) for PCB Carrier Board

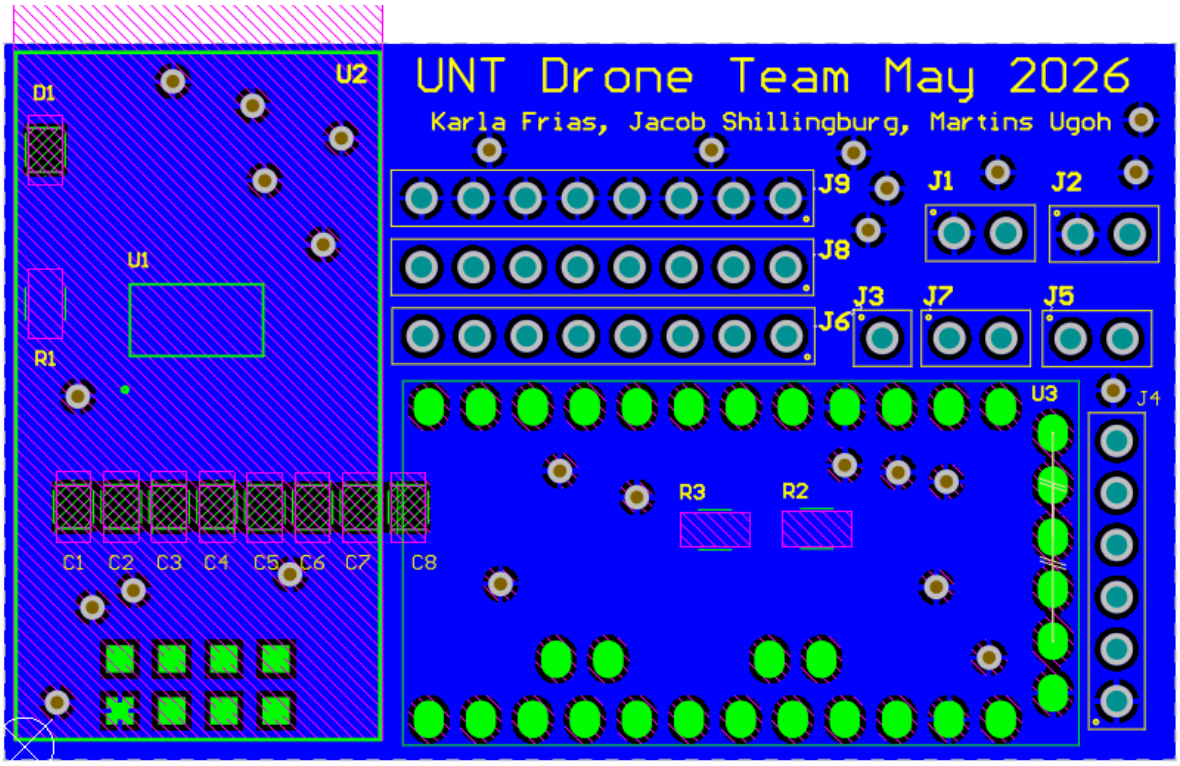


Figure 8: Altium Bottom Layer (GND) for PCB Carrier Board

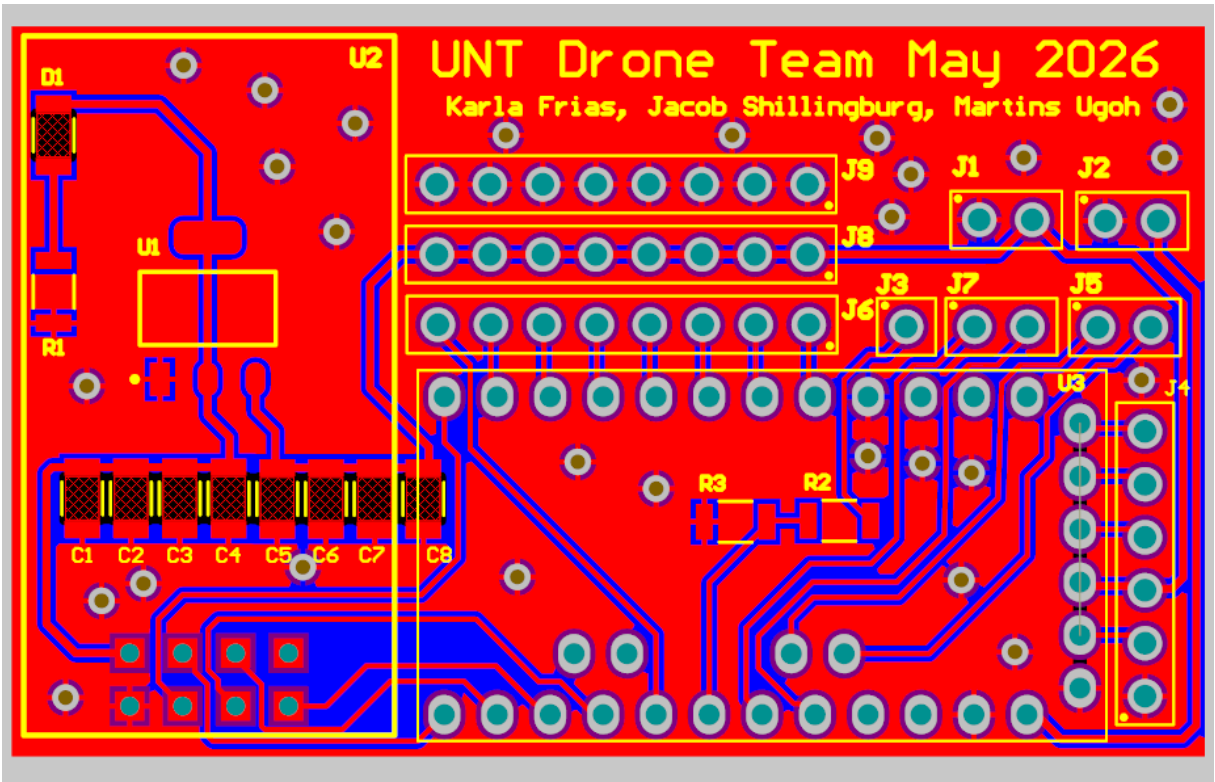


Figure 9: Altium Layout for PCB Carrier Board

Board Assembly

After fabrication, the PCB was assembled and soldered. During assembly, component placement and solder joints were checked carefully to reduce the risk of shorts, poor connections, or incorrect orientation. A high-quality assembly process was important because later debugging becomes significantly more difficult when hardware faults are introduced at this stage.

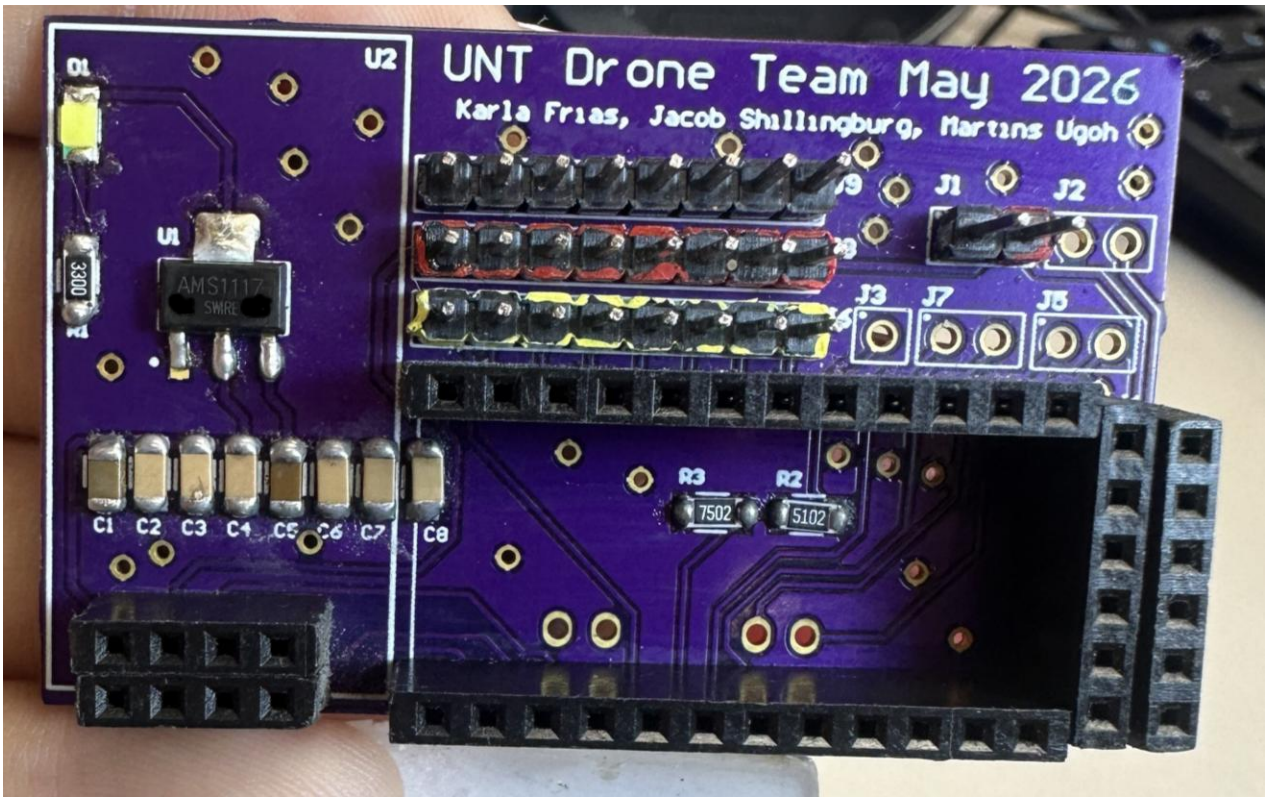


Figure 10: Assembled PCB Carrier Board (Front)

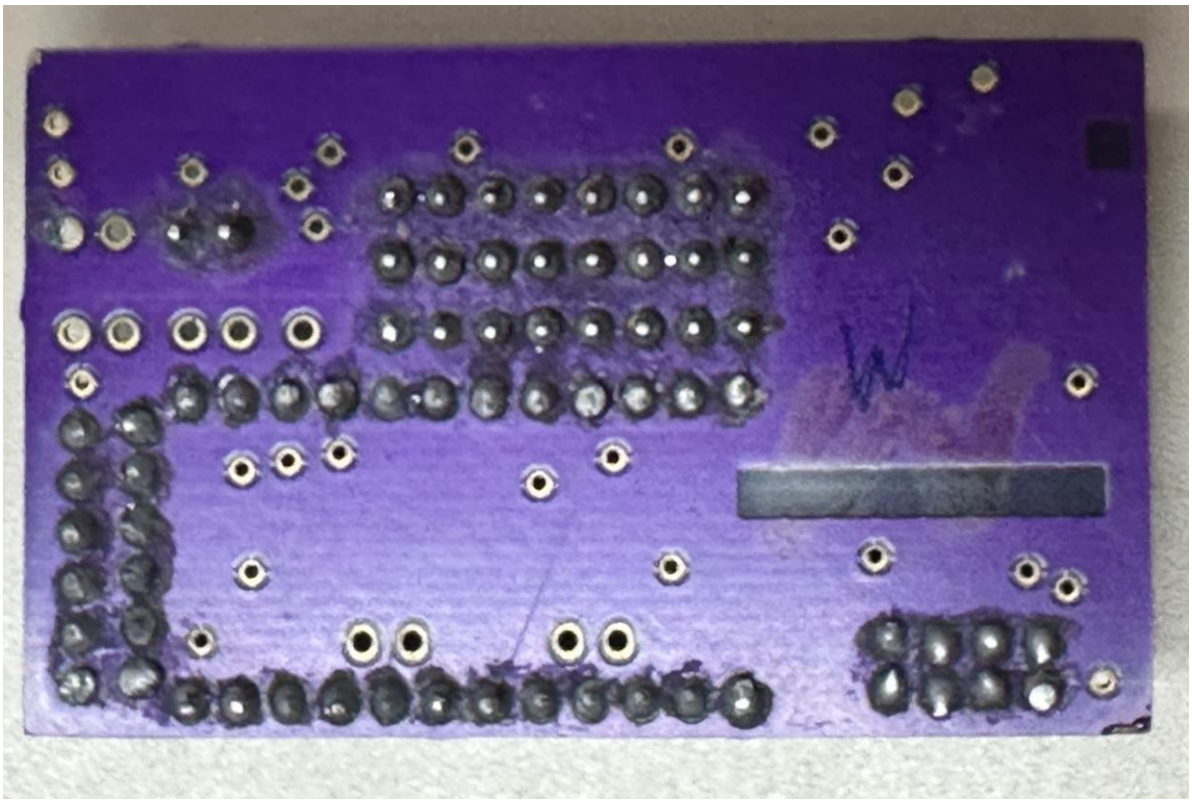


Figure 11: Assembled PCB Carrier Board (Back)

Hardware Validation

Following assembly, the board was validated in the laboratory using oscilloscopes and standard test equipment. These tests focused on verifying power rails, signal continuity, and basic board functionality before attempting full system integration.

Initial testing showed that the board provided the required electrical support for continued development. Stable power delivery and functional interface behavior allowed integration efforts to proceed with confidence that the hardware foundation was operating properly.

Figure 12: Digital Multimeter LED and nRF24 Module Voltage

Board VCC	LED Voltage	Active Current (mA)	Idle Current (mA)
5.0	3.265	5.26	2.21
4.8	3.152	4.99	2.19
4.6V	3.054	4.68	2.14
4.4V	2.976	4.32	2.10
4.2V	2.853	4.08	2.04
4V	2.617	4.19	2.01
3.8V	2.502	3.93	1.98
3.6V	2.448	3.49	1.94
3.4	2.434	2.93	1.92

Transmitter Code

transmitter.ino

```
#include <Wire.h>
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>

RF24 radio(9, 10);   // CE, CSN
const byte address[6] = "00001";

struct DataPacket {
  int x;
  int y;
};

DataPacket data;

const int MPU_ADDR = 0x68;

void setup() {
  Serial.begin(115200);
  delay(1000);

  Serial.println("TX boot");

  Wire.begin();

  Wire.beginTransmission(MPU_ADDR);
  Wire.write(0x6B);
  Wire.write(0);
  Wire.endTransmission(true);

  if (!radio.begin()) {
    Serial.println("radio.begin failed");
    while (1);
  }

```

transmitter.ino

```
  radio.setAutoAck(false);
  radio.setPALevel(RF24_PA_MIN);
  radio.setDataRate(RF24_250KBPS);
  radio.setChannel(100);
  radio.openWritingPipe(address);
  radio.stopListening();

  Serial.println("TX ready");
}

void loop() {
  int16_t AcX, AcY, AcZ, Tmp, GyX, GyY, GyZ;

  Wire.beginTransmission(MPU_ADDR);
  Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(MPU_ADDR, 14, true);

  AcX = Wire.read() << 8 | Wire.read();
  AcY = Wire.read() << 8 | Wire.read();
  AcZ = Wire.read() << 8 | Wire.read();
  Tmp = Wire.read() << 8 | Wire.read();
  GyX = Wire.read() << 8 | Wire.read();
  GyY = Wire.read() << 8 | Wire.read();
  GyZ = Wire.read() << 8 | Wire.read();

  data.x = map(AcX, -17000, 17000, 0, 180);
  data.y = map(AcY, -17000, 17000, 180, 0);   // inverted Y

  data.x = constrain(data.x, 0, 180);
  data.y = constrain(data.y, 0, 180);

  bool ok = radio.write(&data, sizeof(data));

  Serial.print("X: ");
```

```
  Serial.print(data.x);
  Serial.print("  Y: ");
  Serial.print(data.y);
  Serial.print("  | ");
  Serial.println(ok ? "SENT" : "FAILED");

  delay(50);
}
```

Figure 13: Transmitter C++ Code

Receiver Code

reciever.ino

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>
#include <Servo.h>

RF24 radio(9, 10);   // CE, CSN
const byte address[6] = "00001";

Servo pan;
Servo tilt;

struct DataPacket {
  int x;
  int y;
};

DataPacket data;

int panPos = 90;
int tiltPos = 90;

const int deadband = 2;
const float alpha = 0.2;

void setup() {
  Serial.begin(115200);
  delay(1000);

  Serial.println("RX boot");

  pan.attach(3);    // D3 = pan
  tilt.attach(5);   // D5 = tilt

  pan.write(panPos);
  tilt.write(tiltPos);

  if (!radio.begin()) {
    Serial.println("radio.begin failed");
    while (1);
  }

  radio.setAutoAck(false);
  radio.setPALevel(RF24_PA_MIN);
```

```
  radio.setDataRate(RF24_250KBPS);
  radio.setChannel(100);
  radio.openReadingPipe(0, address);
  radio.startListening();

  Serial.println("RX ready");
}

void loop() {
  if (radio.available()) {
    radio.read(&data, sizeof(data));

    data.x = constrain(data.x, 0, 180);
    data.y = constrain(data.y, 0, 180);

    if (abs(data.x - panPos) > deadband) {
      panPos = panPos + alpha * (data.x - panPos);
    }

    if (abs(data.y - tiltPos) > deadband) {
      tiltPos = tiltPos + alpha * (data.y - tiltPos);
    }

    pan.write(panPos);
    tilt.write(tiltPos);

    Serial.print("Pan: ");
    Serial.print(panPos);
    Serial.print("  Tilt: ");
    Serial.println(tiltPos);
  }

  delay(15);
}
```

Figure 14: Receiver C++ Code

System Integration

System integration involved combining the electrical, mechanical, communication, and video subsystems into one functional head tracking platform. After the carrier board was designed, fabricated, and soldered, it was connected with the motion sensor, wireless modules, servo system, camera hardware, and video components to verify complete system operation. This stage was necessary to confirm that the individual subsystems could work together as a single embedded control system.

On the transmitter side, the MPU6050 provided motion data to the Pro Mini ATmega328P 5V/16 MHz, where the input was processed into pitch and yaw values. These two control components were then organized into packets and transmitted through the nRF24L01 wireless communication link. On the receiving side, the transmitted data was read and used to control the servo motors responsible for gimbal pan and tilt motion. This confirmed that the sensing, processing, and wireless control chain could be successfully integrated across both sides of the system.

System integration also included the return video path. A digital video feed was selected for the feedback system to support clearer viewing and better testing at longer distances. This allowed live camera output to be returned to the user display, completing the operator feedback loop and making it possible to observe the effect of head movement on remote camera direction in real time.

Mechanical integration was also an important part of this process. The board, camera hardware, and mounting structure were designed in Fusion 360 and assembled using the required support features, mounting holes, and fasteners. This helped ensure that the electronics could be physically secured within the platform and remain stable during handling and testing. Together, these steps demonstrated that the head tracking system could function as a complete integrated prototype and provided a foundation for continued refinement toward smoother control response and future drone-based deployment.

Testing and Results

Testing of the complete head tracking system was performed in stages to verify electrical operation, wireless communication, mechanical integration, and overall system functionality. The

purpose of testing was to confirm that the transmitter board, receiver board, and camera feedback system could operate together as an integrated platform for remote camera control.

Initial testing focused on power regulation and board-level operation. The custom carrier board was powered and checked to verify that the regulator stage provided the required supply levels for the onboard electronics. Voltage measurements and current testing were used to confirm stable operation across the expected input range. The board remained functional as the supply was varied, and current measurements were recorded for both idle and active conditions. These tests helped verify that the PCB could reliably support the microcontroller, sensor, and wireless transceiver.

Additional testing was then performed on the sensing and control path. The MPU6050 sensor successfully provided motion-related input to the Pro Mini ATmega328P, allowing head motion data to be processed and separated into pitch and yaw control components. These two data values were then organized for packet-based transmission through the nRF24L01 wireless link. On the receiving side, the transmitted data was read and used to drive the servo system. This confirmed that the end-to-end control path, from sensed head motion to servo output, was functioning as intended.

System testing also showed that the head tracking direction was operating correctly and no longer required axis inversion. However, the motion response can still be improved to make the tracking smoother. Although the system followed the expected directional behavior, further tuning would help reduce abrupt motion and produce more natural camera movement.

The return video path was also tested as part of the full system evaluation. A digital video feed was selected for the return path in order to support testing at longer distances and provide clearer visual feedback during operation. This allowed the camera output to be viewed more effectively during system testing and confirmed that the visual feedback path was functional. Establishing a working return video link was an important result because it demonstrated that the operator could receive real-time visual information while using the head tracking system.

Mechanical integration was evaluated through physical assembly and fit testing of the carrier board, camera hardware, and mounting structure. CAD models developed in Fusion 360 supported the placement of these components before fabrication, and the assembled hardware confirmed that the board and related components could be mounted within the intended platform. Fastening methods, alignment, and structural support were reviewed during assembly to confirm that the hardware could remain secured during handling and testing.

Testing of the larger platform identified system-level limitations as well. While the PCB itself was compact and lightweight enough for integration, one of the heavier drone configurations limited the ability to perform multiple full test flights. As a result, complete flight-based validation of the entire system was restricted. Even so, subsystem testing confirmed that the electronics, communication link, servo output, and video return path were operating and ready for further refinement.

Overall, the testing results demonstrated that the project successfully produced a functional prototype head tracking system. The PCB operated correctly, packet-based wireless communication was established, servo control based on pitch and yaw data was demonstrated, and

the digital return video path was successfully connected. These results show that the system provides a solid foundation for continued development toward smoother tracking performance, stronger connector reliability, environmental protection, and future drone or first responder deployment.

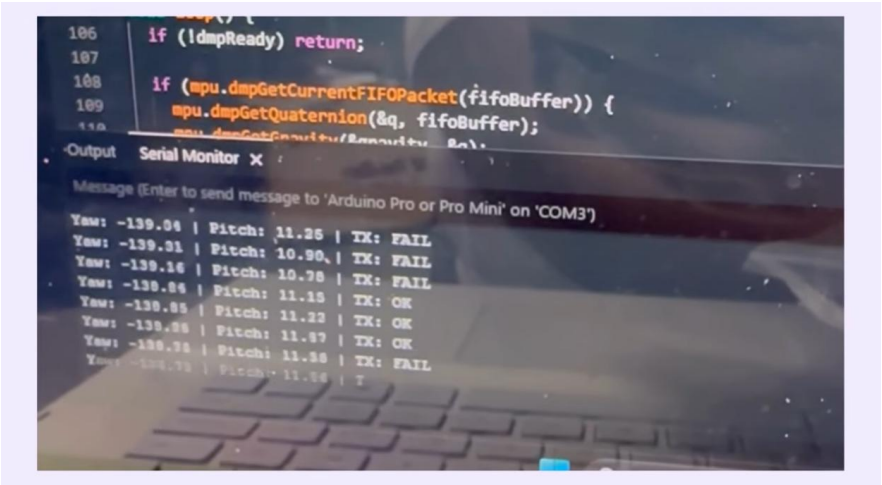


Figure 15: Serial Monitor Output During Testing (test 1)

Figure 16: Pan/Tilt Output During Testing (test 1)

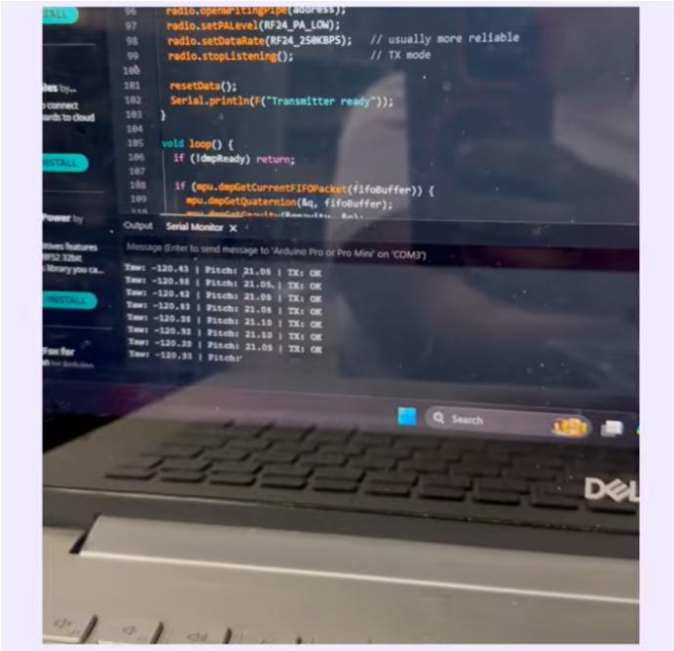


Figure 17: Serial Monitor Output During Testing (test 2)

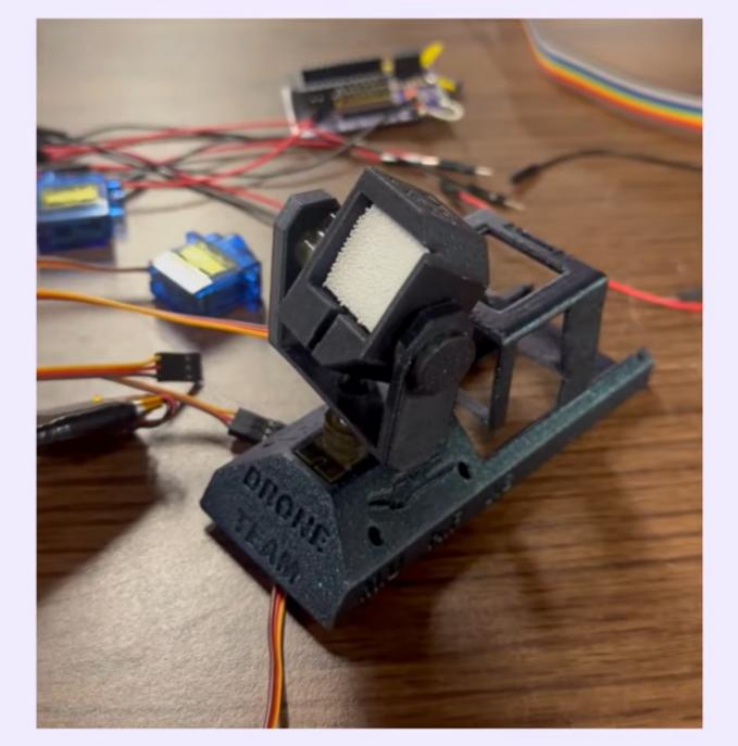


Figure 18: Pan/Tilt Output During Testing (test 2)

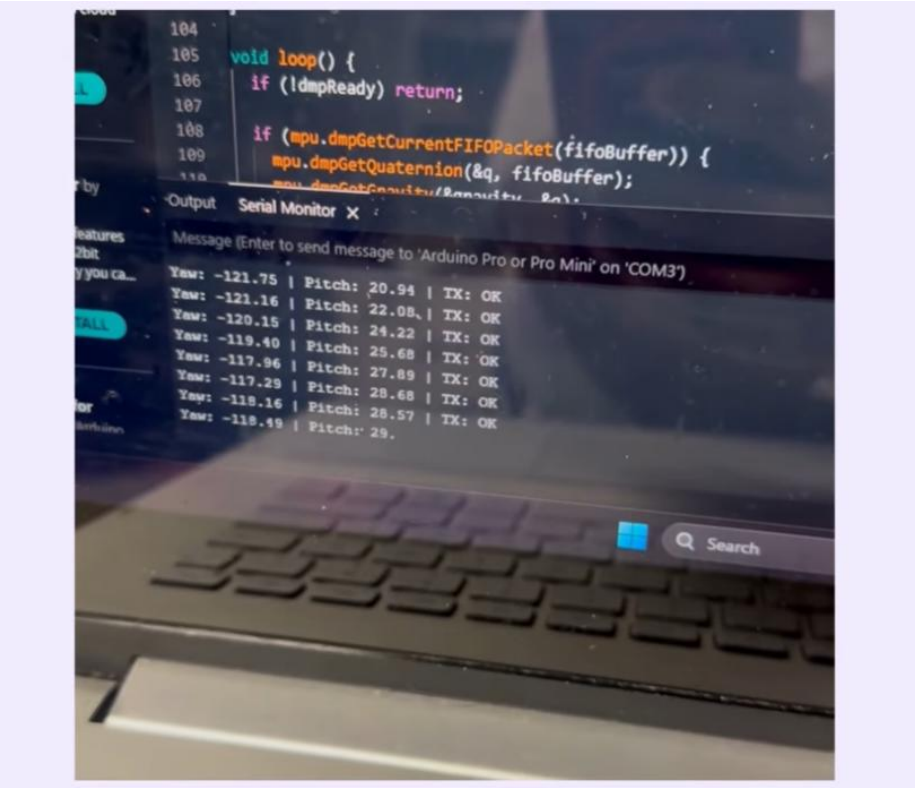


Figure 19: Serial Monitor Output During Testing (test 3)

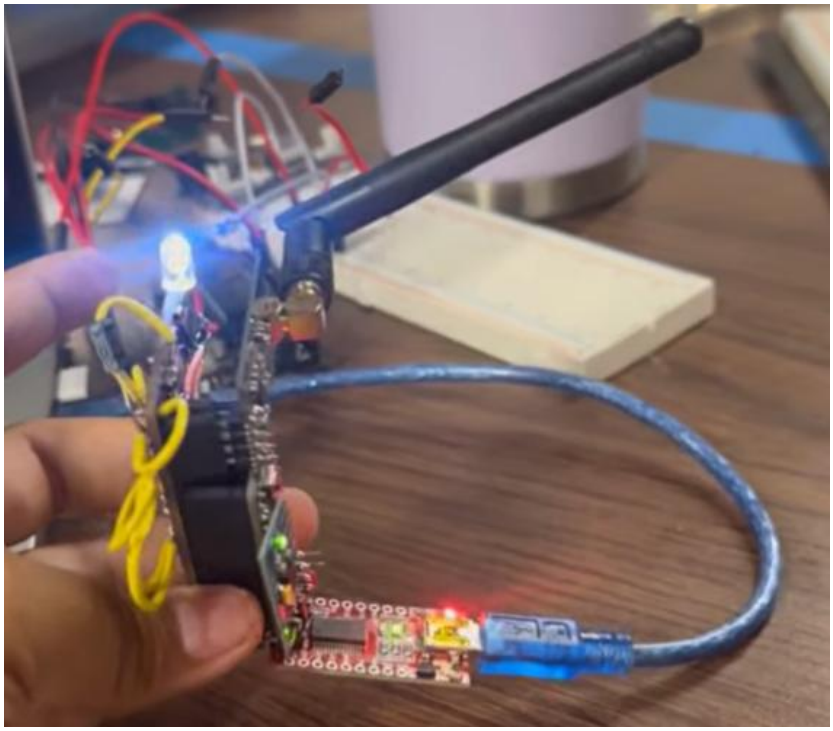


Figure 20: Transmitter PCB During Testing (test 3)

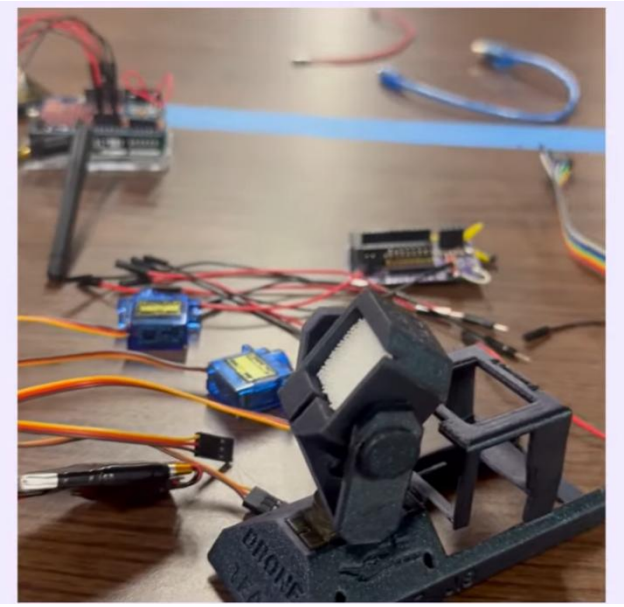


Figure 21: Pan/Tilt Output During Testing (test 3)

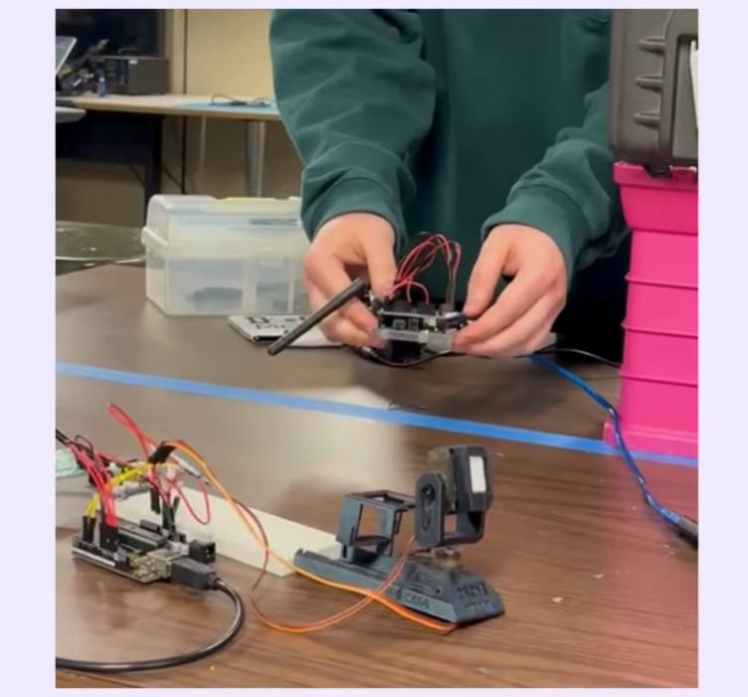


Figure 22: Pan/Tilt Output During Testing entire system (test 4)

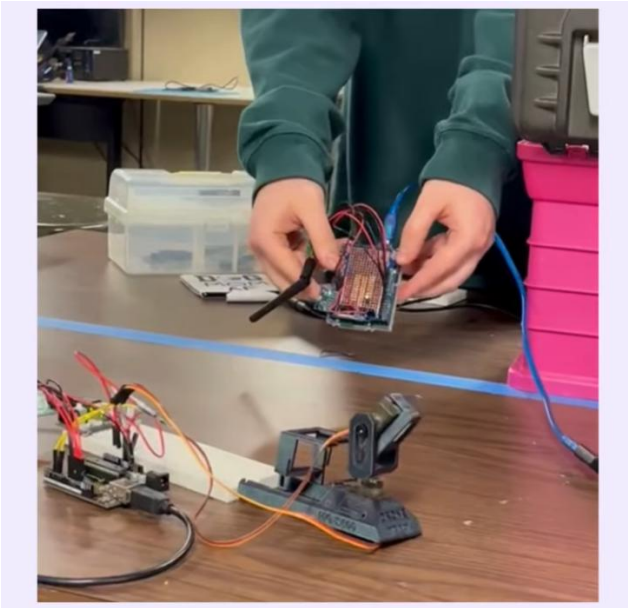


Figure 23: Pan/Tilt Output During Testing Entire System (test 5)

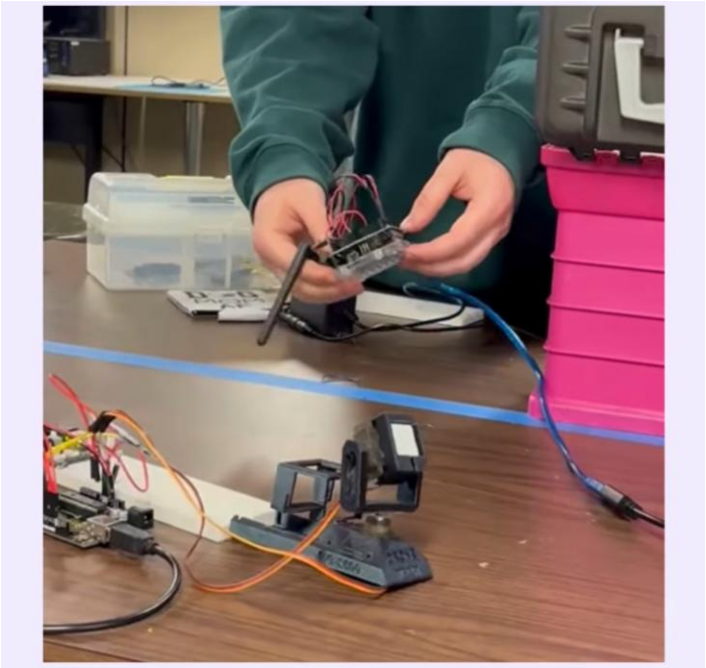


Figure 24: Pan/Tilt Output During Testing Entire System (test 6)

Challenges and Discussion

Several challenges were encountered during the design, assembly, and integration of the head tracking system. One challenge was related to the overall drone platform used during development. While the PCB itself was lightweight and suitable for integration, the heavier drone design limited the ability to perform multiple test flights. This reduced opportunities for full flight-based validation and made it more difficult to evaluate the system under repeated real operating conditions.

Another challenge involved achieving smooth and stable camera response from the head tracking input. Although the system functioned correctly and the tracking direction no longer required axis inversion, the motion response can still be improved to provide smoother operation. This highlighted the importance of control tuning and system refinement in addition to basic hardware functionality.

A further challenge involved the physical connections used on the board. The current pin headers allowed the system to be assembled and tested, but they are not ideal for repeated use or long-term deployment because they can become loose or stripped. This created concerns about reliability during testing and future drone integration. As a result, connector durability became an important design consideration beyond the initial electrical functionality.

System-level factors such as platform limitations, electrical design, mechanical integration, connector reliability, and environmental protection all contribute to final performance. The work completed in this project established a functional prototype and also revealed the practical improvements needed to make the system smoother, more durable, and better suited for real-world drone and first responder deployment.

Conclusion

The head tracking system developed in this project successfully demonstrated the design and integration of a custom PCB for remote camera control applications. The project included schematic capture, PCB layout in Altium, component selection, bill of materials development, soldering, hardware validation, and integration with the camera and headset system. A clear digital camera feed was successfully connected to the headset, confirming that the system established a functional visual link and a solid foundation for directional camera control.

The final prototype showed that the head tracking system operates correctly and no longer requires axis inversion, although additional refinement is still needed to make the motion response smoother. The project also identified practical improvements for long-term reliability, including replacing the current pin headers with more secure connectors and applying conformal coating to protect the board in drone-based environments.

Overall, this work demonstrated the feasibility of a compact, PCB-based head tracking platform for remote sensing and camera positioning. With continued refinement, the system can support more reliable operation in drone and first responder applications, where intuitive control, stable visual feedback, and durable hardware are especially important.

Future Work

Future work will focus on improving the overall performance, durability, and field readiness of the head tracking system. Although the PCB was successfully designed, assembled, and integrated with the camera display, additional refinement is still needed to improve the smoothness of the head tracking response. The current system functions as intended, but further tuning of the control behavior would help produce more stable and natural camera motion during operation.

Future development should focus on optimizing responsiveness and reducing any jitter or abrupt movement in the control output. This would improve user experience and make the system more suitable for practical remote viewing applications.

Another important area for future work is the replacement of the current pin headers with a more secure connection method. The existing pin headers can be easily stripped or loosened during repeated use, which may reduce reliability during testing and deployment. Replacing them with a stronger and more durable connector solution would improve the robustness of the system and support long-term integration.

Because the system is intended for use on a drone platform, future work should also include applying conformal coating to the PCB. Conformal coating would help protect the electronics from moisture, dust, contamination, and other environmental conditions encountered during flight operations. This added protection would improve reliability and make the board more suitable for outdoor and field deployment.

These improvements would support the development of a more reliable system for first responder applications, where stable remote vision, durability, and rapid situational awareness are especially important. With additional refinement, the system could provide a stronger platform for drone-based camera control in emergency response and other field environments.

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Appendix

Appendix A: 3D Model Files

https://github.com/karlafrias/EE-Senior-Design-Head-Tracking-Gimbal-System/tree/main/3D%20Models

Appendix B: PCB Files

https://github.com/karlafrias/EE-Senior-Design-Head-Tracking-Gimbal-System/tree/main/PCB%20Files

Appendix C: PCB Component List

Transmitter/Receiver PCB Parts List				
Label	Description	Value	Package	Required?(no= optional)
D1	LED White	n/a	1206 SMD	Yes
R1	1/8W 1% Resistor	330 Ω	1206 SMD	Yes
U1	AMS1117 – 3.3V	n/a	SOT-223	Yes
C1	Bulk Decoupling Voltage Regulator Capacitor	22uF	1206 SMD	Yes
C2	Decoupling Voltage Regulator Capacitor	.1uF	1206 SMD	Yes
C3	Decoupling Voltage Regulator Capacitor	.1uF	1206 SMD	No
C4	Decoupling Voltage Regulator Capacitor	.1uF	1206 SMD	No
C5	Bulk Decoupling Voltage Regulator Capacitor	10uF	1206 SMD	Yes
C6	Decoupling Voltage Regulator Capacitor	.1uF	1206 SMD	Yes
C7	Decoupling Voltage Regulator Capacitor	.1uF	1206 SMD	No
C8	Decoupling Voltage Regulator Capacitor	.1uF	1206 SMD	No
U2	E01-ML01DP5 Wireless Transmission Module	n/a	n/a	Yes
R2	1/8W 1% Resistor	51 Ω	1206 SMD	No

R3	1/8W 1% Resistor	75 Ω	1206 SMD	No
U3	Pro Mini Atmega328P 5V/16MHz	n/a	n/a	Yes
J1	2.54MM Male Header (VCC)	n/a	2 pin	Yes
J2	2.54MM Male Header	n/a	2 pin	No
J3	2.54MM Male Header	n/a	1 pin	No
J4	2.54MM Female Header	n/a	6 pin	Yes
J5	2.54MM Male Header	n/a	2 pin	No
J6	2.54MM Male Header (Signal Rail)	n/a	8 pin	Yes
J7	2.54MM Male Header	n/a	2 pin	No
J8	2.54MM Male Header (5V Rail)	n/a	8 pin	Yes
J9	2.54MM Male Header (Ground)	n/a	8 pin	Yes

Table : Transmitter/Receiver Component List

Transmitter PCB Parts List			
Label	Description	Placement	Required?
n/a	MPU6050 Accelerometer/Gyroscope Sensor	Attaches to Arduino	Yes

Table : Transmitter Component List

Appendix D: System Component List

System Parts List		
Name	Description	Required?
Walksnail Avatar FPV VRX With Moonlight Kit	FPV camera kit for drone use	Yes
Walksnail Avatar HD Goggles L	Digital headset	Yes
FEELWORLD P6XL 6 inch	Monitor	Yes
18650 3000mAh Battery Cell	Receiver board battery	Yes

3.7V LiPo Battery 500mAh	Transmitter board battery	Yes
5V 1A Boost Power Supply Module	Use in parralel with battery packs for charging and 5V boost for PCB board	Yes

Table : System Component List

Appendix E: Transmitter/Receiver Code

https://github.com/karlafrias/EE-Senior-Design-Head-Tracking-Gimbal-System/tree/main/Code

Appendix F: Component Data Sheets

https://github.com/karlafrias/EE-Senior-Design-Head-Tracking-Gimbal-System/tree/main/Datasheets

Appendix G: YouTube Video

https://www.youtube.com/watch?v=_BPLgMK_N9A